

Forecasting Brazilian Ethanol Prices

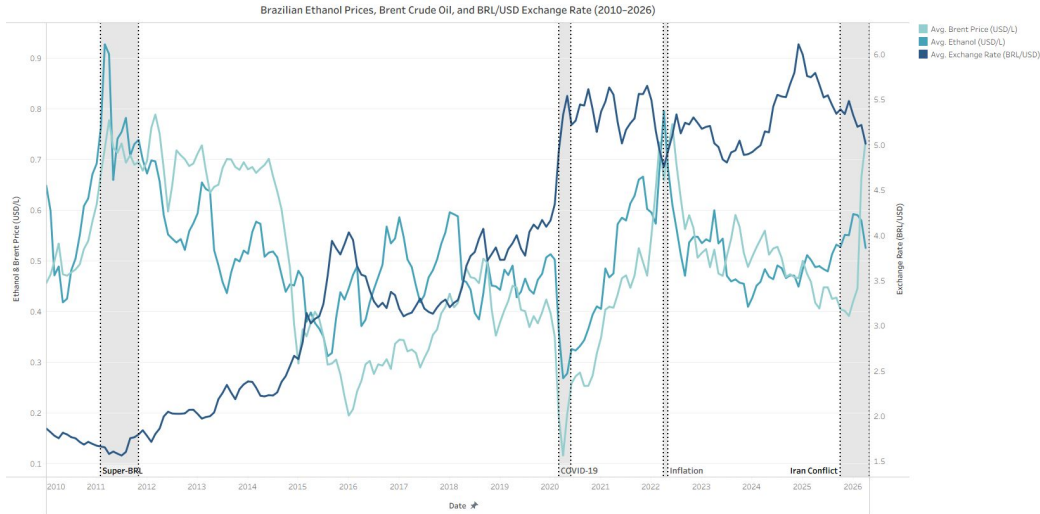
Group 3

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Predicting Brazilian Ethanol Prices



Observed market history: ethanol, Brent, and BRL/USD move through multiple regimes, so static correlation alone is not enough.

The 2026 oil shock created volatility in global fuel markets, but Brazil has a partial buffer through its flex-fuel ethanol system.

Our goal is to forecast weekly Paulínia ethanol prices and identify which external drivers improve prediction, not only explanation.

The final output should support short-horizon decisions for supply planning, retail pricing, inflation monitoring, and subsidy planning.

Research Question

Which factors contribute most to explaining weekly variation in Brazilian ethanol prices?

Candidate Drivers Tested

- 📈 Brent crude oil prices (Global)
- 🏦 BRL/USD exchange rate
- 🌾 Harvest seasonality (Local)
- ⚠️ 2026 geopolitical shock

H1a: Brent crude oil price is a dominant predictor of Brazilian ethanol prices.

H2a: The BRL/USD exchange rate is a significant mediator of how oil shocks translate into domestic ethanol prices.

H3a: Sugarcane harvest seasonality explains additional variance beyond macroeconomic factors.

H4a: The 2026 geopolitical shock represents a structural break where the relationship between oil and ethanol intensifies.

Framework: Incremental **ARIMAX** modeling with expanding-window Times Series Cross-Validation

Metrics: RMSE, AIC, Diebold-Mariano test

Design & Data Collection Methods

Source	Variable
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EIA
Brent price USD/L
(independent)

CEPEA*
Ethanol price USD/L
(dependent)

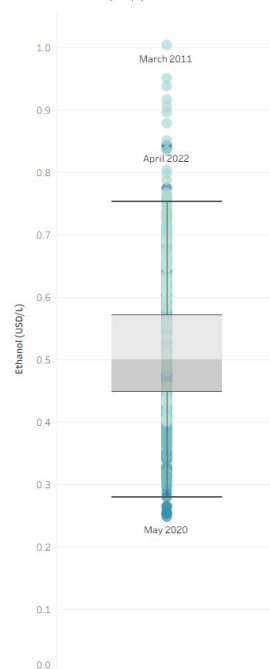
FRED
BRL/USD exchange rate
(independent)

*February 2010 to April 2026
Daily → weekly average
848 final observations

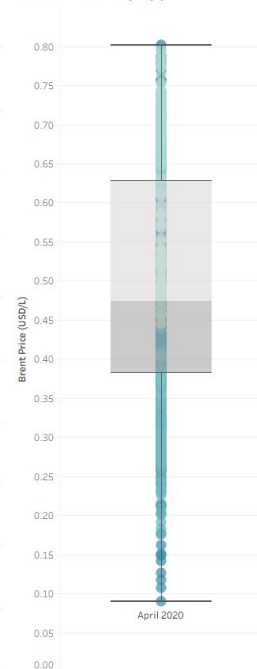
Engineered variables

- HARVEST_DUMMY (Apr-Nov)
- CONFLICT_DUMMY (Oct 2025 to Apr 2026)
- DUMMY_2022_INFLATION
- DUMMY_2011_CRISIS
- DUMMY_2020_COVID

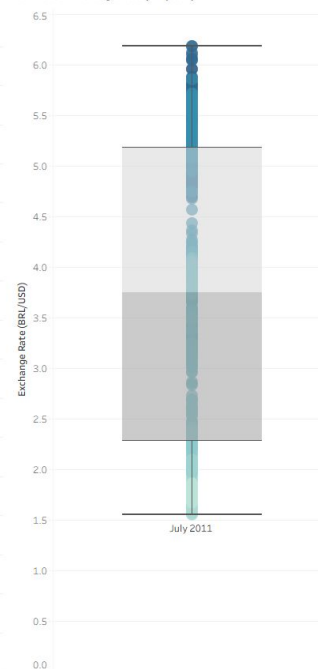
Box Plot for Ethanol (USD/L)



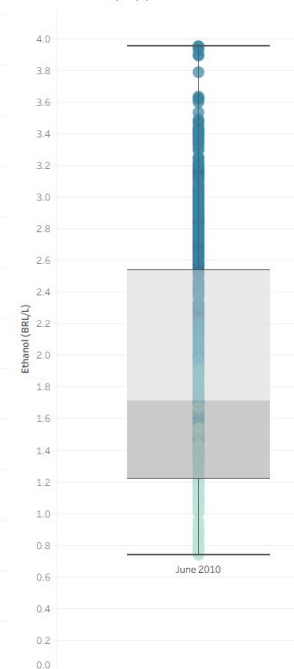
Box Plot for Brent Price (USD/L)



Box Plot for Exchange Rate (BRL/USD)



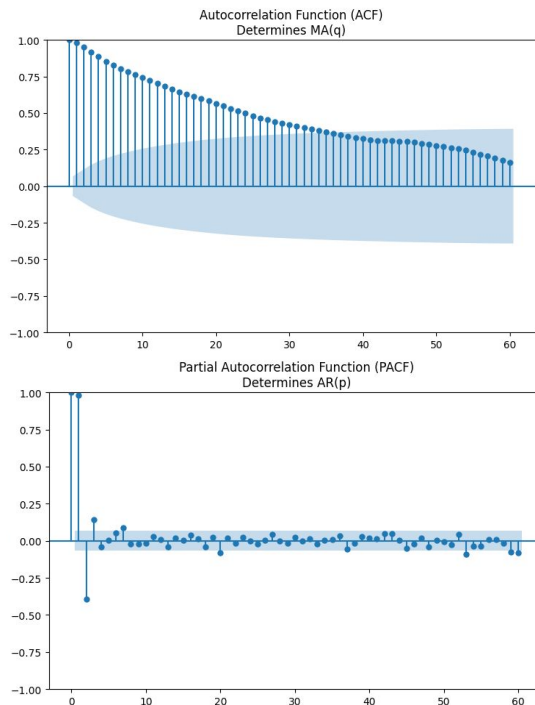
Box Plot for Ethanol (BRL/L)



Missing values were handled by LOCF to avoid look-ahead bias.
Historical outliers were retained and modeled through control intervention dummies.

Methodology

Traditional time-series diagnostics + systematic Grid Search + Time Series Cross-Validation + Robustness Check



Stationary Testing

ADF p-value 0.017 $\rightarrow$ $d=0$

ACF Test

$q=0$

PACF Test

$p=3$

Train-test Split

Chronological 80/20

ARIMAX Model

Preliminar baseline

Validation

Grid Search + CV

Robustness Check

Sensitivity & Diagnostics

Model Building

Model	Key variable	AIC	Test RMSE	Interpretation	Hypothesis Evaluation
Baseline	Ethanol history + controls	-3425.2527	0.0588	History predicts ethanol	————
Model 2 	Lagged Brent	-3421.3766	0.0575	Improves forecasts	H1a Supported
Model 3	Lagged exchange rate	-3442.2246	0.1328	Explains past; unstable forecast	H2a Supported (Explanatory perspective)
Model 4	Harvest dummy	-3419.6672	0.0577	No added value	H3a Rejected
Model 5	Conflict dummy	-3419.3767	0.0577	No added value	H4a Rejected*

*H4 was later supported only in the isolated wartime scenario analysis, not as a general forecasting specification.

All results are statistically significant with Diebold-Mariano test with p-value < 5%

Geopolitical Shock Testing - Scenario Simulation

The 2026 conflict behaves more like a change in sensitivity than a flat price premium.

Simulated Scenarios

```
--- Testing H4: Additive vs. Multiplicative Structural Break ---  
Baseline (Peacetime Rules) RMSE: 0.0575  
  
--- Additive Simulations (Flat Premium) ---  
Scenario Add-A (+$0.05) RMSE: 0.0947  
Scenario Add-B (+$0.10) RMSE: 0.1401  
Scenario Add-C (+$0.20) RMSE: 0.2366  
  
--- Multiplicative Simulations (Oil Sensitivity) ---  
Scenario Mult-A (1.3x) RMSE: 0.0630  
Scenario Mult-B (1.6x) RMSE: 0.0693  
Scenario Mult-C (2.0x) RMSE: 0.0784
```

Isolated Wartime Forecast

Scenario	RMSE	Performance vs Baseline
Baseline Brent-only	0.0473	Reference
Additive (+\$0.20 flat)	0.1671	✗ 253% worse
Multiplicative (2.0x)	0.0353	✓ 25% better

During crisis weeks, decision-makers should stress-test the Brent sensitivity rather than manually add a constant conflict premium.

Grid Search & Cross-Validation

Top 5 Model Configuration

Rank	(p, d, q)	Average CV RMSE	Std CV RMSE	Average AIC	Average BIC
1	(0, 0, 0)	0.091960	0.023593	-785.456106	-763.279511
2	(0, 0, 1)	0.093469	0.023937	-1131.579682	-1105.706987
3	(0, 0, 2)	0.094810	0.022288	-1345.977681	-1316.408887
4	(0, 1, 0)	0.098400	0.026180	-1617.037513	-1598.576895
5	(0, 0, 3)	0.100788	0.025605	-1471.761087	-1438.496194

- 50 ARIMAX order combinations
 - p [0, 1, 2, 3, 4]
 - d [0, 1]
 - q [0, 1, 2, 3, 4]
- CV with expanding-window across multiple historical forecasting windows

Holdout test evaluation

Model	RMSE
Baseline ARIMAX(3, 0, 0)	0.055620
CV-Selected ARIMAX(0, 0, 0)	0.057839

The baseline Brent specification remained the strongest economic model for forecasting ethanol.

Diagnostics & Robustness

The selected model is useful but not perfect, especially under extreme volatility.

Robustness check

2-week & 4-week forecast horizon

```
--- True (Rolling-Origin) Robustness Check (Multi-Week Horizons) ---
```

```
Evaluating a True Rolling-Origin 2-Week Ahead Forecast Horizon:
```

```
-> True Rolling Naive RMSE: 0.0213
```

```
-> Model 1 (Pure AR(3) Rolling) RMSE: 0.0570
```

```
-> Model 2 (Brent Rolling) RMSE: 0.0198
```

```
SUCCESS: Model 2 maintains structural predictive dominance at horizon t+2!
```

```
Evaluating a True Rolling-Origin 4-Week Ahead Forecast Horizon:
```

```
-> True Rolling Naive RMSE: 0.0310
```

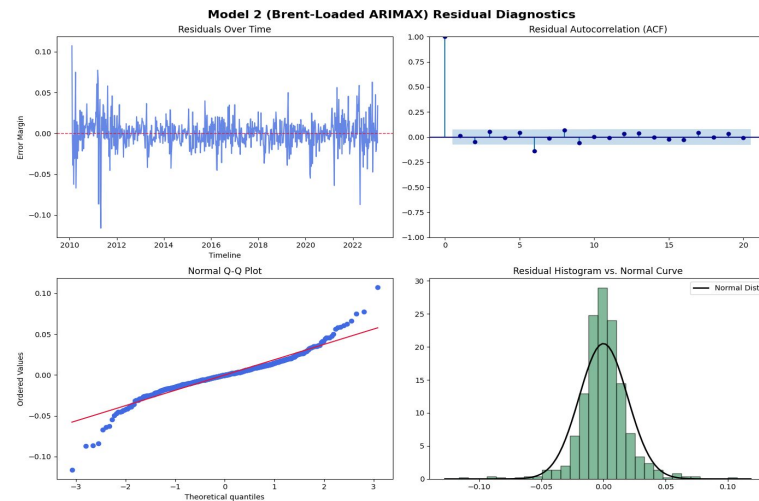
```
-> Model 1 (Pure AR(3) Rolling) RMSE: 0.0559
```

```
-> Model 2 (Brent Rolling) RMSE: 0.0293
```

```
SUCCESS: Model 2 maintains structural predictive dominance at horizon t+4!
```

- Brent remains the most stable general predictor.
- Longer forecast horizons should be interpreted as planning signals, not precise point predictions.

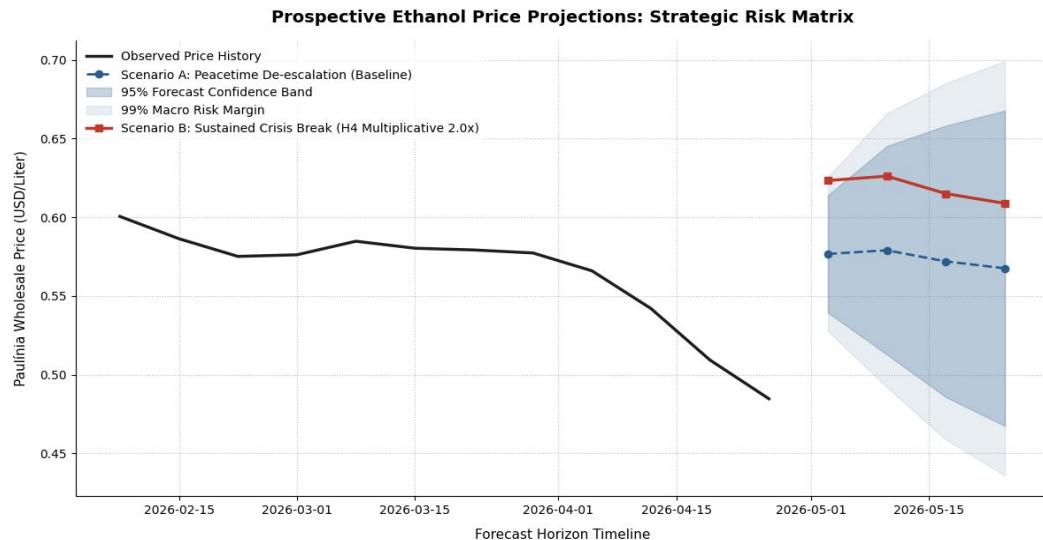
Residual Diagnostic



- Residuals are mostly centered around zero, so the model is not consistently over- or under-forecasting.
- Ljung-Box $p=0.0077$ warns that some autocorrelation remains.
- Heavy tails mean forecasts require caution during shock periods.

Final Forecast Projection

Four-week prospective forecast supports short-horizon planning under alternative market scenarios.



Normal de-escalation: prices are expected to stabilize or ease as pressure from Brent relaxes.

Prolonged crisis: prices remain elevated and uncertainty widens, making inventory timing more important.

Operational use: convert weekly projections into purchase, delivery, pricing, and monitoring decisions.

Business Impact & Recommendations

The same forecast can be translated into different decisions for different stakeholders.

Petrobras & distributors

Plan ethanol delivery volumes and regional allocation before wholesale price changes.

Gas station operators

Time wholesale purchases, protect margins, and prepare pricing near the 70% ethanol/gasoline threshold.

Central Bank of Brazil

Aggregate weekly forecasts into monthly fuel indicators for inflation monitoring.

Ministry of Finance

Plan subsidy exposure and separate temporary spikes from persistent structural increases.

Impact logic: better fuel-price anticipation reduces operational surprise, supports more disciplined pricing, and improves policy timing.

Recommendations:

- Use Brent model for routine short-horizon forecasting.
- Use multiplicative Brent-sensitivity scenarios during geopolitical stress.
- Future work: regional models, diesel channel, GARCH volatility, and regime-switching models.

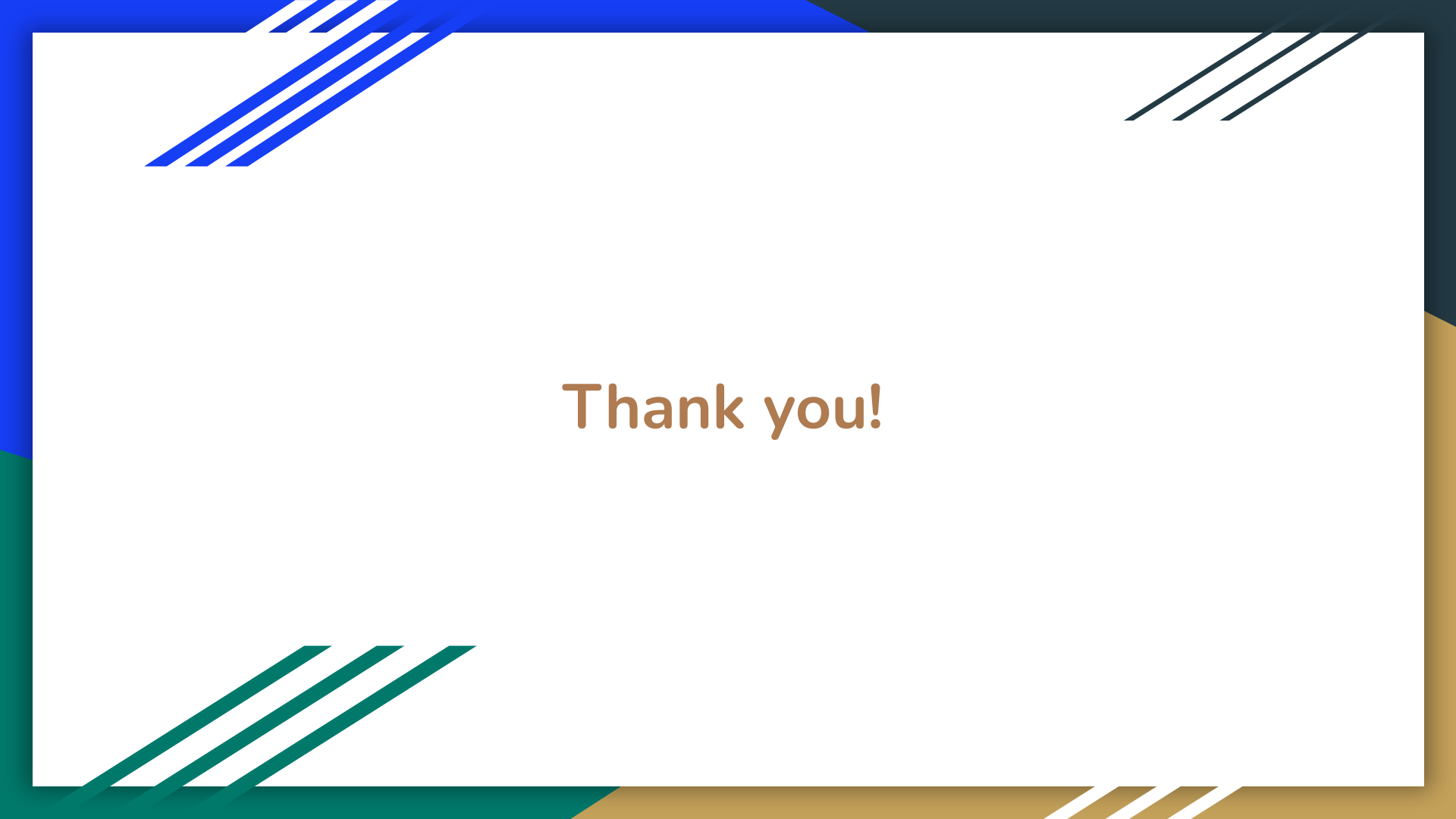
Limitations & Conclusion

Limitations

- Limited Naive Benchmarking
- Single regional ethanol price series (Paulínia)
- Residual autocorrelation remains
- Geopolitical event is unique
- Exchange rate forecasting instability
- Cross-Validation and Holdout divergence

Conclusion

- The Brent-based ARIMAX model is the strongest general forecasting specification.
- Exchange rate and harvest seasonality are economically meaningful but do not improve future prediction in this model.
- The 2026 shock is better handled through scenario stress-testing than a simple additive dummy.



Thank you!